

Homework Assignment #2

Assigned: 12 February 2026

Due: 3 March 2026

Reading (from Papoulis and Pillai):

Chapter 5, sections 5-1 – 5-2; Chapter 6, sections 6-1, 6-4, and 6-6 – 6-7; Chapter 7, sections 7-1 – 7-2

Work to hand in:

1. Problem 5-35, page 166.
2. The RV $\mathbf{n}$ is normal with mean zero and variance one. Under H_0 , $\mathbf{x} = \mathbf{n}$. Under H_1 , $\mathbf{x} = s + \mathbf{n}$ where $s \geq 0$.
 - (a) If $P(H_0) = P(H_1)$, for what range of x should a hypothesis test decide in favor of H_1 ? In terms of s , what are P_d , P_f , and P_e for this test?
 - (b) Take $s = 1$ and find P_d and P_f in terms of the ratio $T = P(H_0)/P(H_1)$. Use these expressions to sketch the ROC curve.

3. Under H_0 , $\mathbf{x}$ has conditional PDF $f_{\mathbf{x}}(x|H_0)$. Under H_1 its conditional PDF is $f_{\mathbf{x}}(x|H_1)$. Show that the Bayesian decision rule

$$\rightarrow H_1 = \left\{ \frac{f_{\mathbf{x}}(x|H_1)}{f_{\mathbf{x}}(x|H_0)} > \frac{P(H_0)}{P(H_1)} \right\}$$

minimizes the probability of error P_e .

4. The joint PDF of $\mathbf{x}_1$ and $\mathbf{x}_2$ is

$$f_{\mathbf{x}_1, \mathbf{x}_2}(x_1, x_2) = \begin{cases} a & \text{if } x_1 \geq 0, x_2 \geq 0, \text{ and } x_1 + x_2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of a .
 - (b) Calculate $E[\mathbf{x}_1]$, $\text{var}(\mathbf{x}_1)$, and $\text{cov}(\mathbf{x}_1, \mathbf{x}_2)$.
5. Problem 5-12, page 165, parts (a) and (b).
6. Problem 5-17, page 165.
7. Problem 7-11, page 300.
8. Problem 6-12, page 236.
9. Problem 6-20, page 237, parts (a) and (b).
10. Problem 7-14, page 301.
11. The joint PDF of $\mathbf{x}_1, \dots, \mathbf{x}_n$ is

$$f_{\mathbf{x}}(X) = \begin{cases} a & \text{if } x_k \geq 0 \text{ for } k = 1, \dots, n \text{ and } x_1 + \dots + x_n \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of a .
 - (b) Calculate $E[\mathbf{x}_k]$, $\text{var}(\mathbf{x}_k)$, and the covariance matrix $C_{\mathbf{X}}$.

12. The joint PDF of n non-negative RVs $\mathbf{x}_1, \dots, \mathbf{x}_n$ has the form

$$f_{\mathbf{x}}(x_1, \dots, x_n) = p(x_1 + \dots + x_n) \quad x_k \geq 0 \text{ for } k = 1, \dots, n$$

where p is a non-negative function.

- (a) Find a condition on p so that the PDF is properly normalized.
- (b) With $N > n$, verify that this condition is satisfied for

$$f_{\mathbf{x}}(X) = \begin{cases} \frac{N!}{(N-n)!} (1 - x_1 - \dots - x_n)^{N-n} & \text{if } x_k \geq 0 \text{ for } k = 1, \dots, n \text{ and } 0 \leq x_1 + \dots + x_n \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

13. Suppose the natural logarithm of the characteristic function of $\mathbf{x}$ is expanded in a power series around $\omega = 0$; i.e.,

$$\log \Phi_{\mathbf{x}}(\omega) = \sum_{k=1}^{\infty} \beta_k \frac{\omega^k}{k!}$$

The coefficients β_k are the *cumulants* of $\mathbf{x}$.

- (a) Show that $\beta_1 = (-i)\mathbf{E}[\mathbf{x}]$ and $\beta_2 = (-i)^2 \text{var}(\mathbf{x})$.
 - (b) If $\mathbf{x}_1, \dots, \mathbf{x}_n$ are independent and $\mathbf{z} = \mathbf{x}_1 + \dots + \mathbf{x}_n$, show that the k^{th} cumulant of $\mathbf{z}$ is the sum of the k^{th} cumulants of the $\mathbf{x}_j$.
14. Problem 4-21, page 120.
15. (Computer problem) RVs $\mathbf{x}_1, \dots, \mathbf{x}_4$ are iid and uniformly distributed on $[-1, 1]$.
- (a) Generate 10,000 independently realized data values for each of these RVs and plot a histogram for $\mathbf{x}_1$.
 - (b) Define $\mathbf{y}_1 = \mathbf{x}_1 + \mathbf{x}_2$ and $\mathbf{y}_2 = \mathbf{x}_3 + \mathbf{x}_4$. Show that $\mathbf{y}_1$ and $\mathbf{y}_2$ are independent and find their PDFs. Add the $\mathbf{x}_1$ and $\mathbf{x}_2$ data to produce 10,000 data for $\mathbf{y}_1$. Similarly, add the $\mathbf{x}_3$ and $\mathbf{x}_4$ data to produce 10,000 data for $\mathbf{y}_2$. Plot a histogram for $\mathbf{y}_1$.
 - (c) Define $\mathbf{z} = \mathbf{y}_1 + \mathbf{y}_2$. Find the PDF, mean, and variance of $\mathbf{z}$. Add the $\mathbf{y}_1$ and $\mathbf{y}_2$ data to produce 10,000 data for $\mathbf{z}$ and plot a histogram for $\mathbf{z}$.
 - (d) Generate 10,000 data from a normal PDF with the same mean and variance as $\mathbf{z}$. Perform a two-sample Kolmogorov-Smirnov goodness of fit test between this data and the $\mathbf{z}$ data and interpret the result.